UNITED STATES INTERNATIONAL UNIVERSITY – AFRICA

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Nairobi Cost of Living Dashboard (2019–2024): Area-Level Expense Trends and Budget Insights

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Part A: Data Acquisition and Understanding

Introduction

This report presents a Power BI dashboard developed to analyse the monthly cost of living across 20+ residential areas in Nairobi, Kenya, from 2019 to 2024. The dataset includes five expense categories: rent, food, transport, utilities, and miscellaneous costs. The primary business problem is that individuals, employers, and policymakers lack a consolidated, time-intelligent view of living expenses by area. Without this, relocation planning, salary benchmarking, and inflation tracking remain inefficient.

The dashboard addresses this problem by enabling interactive filtering by year, area, expense type, and area tier (premium, middle, affordable, satellite). Key measures include total expense, year-to-date spending, month-over-month growth, budget variance, and category contribution. The model follows a star schema with one fact table and five dimension tables, built entirely in Power BI using Power Query transformations and DAX measures.

The analysis reveals that rent is the dominant cost driver, area selection (e.g., Githurai vs. Karen) is the strongest lever for cost control, and seasonal spikes occur in November and December. Based on these insights, three actionable recommendations are provided: area-based relocation guidance, tiered salary zones for employers, and targeted cost reduction priorities for households.

1. Domain and Business Problem

Domain: Urban economics / cost of living analysis (real estate, personal finance, salary benchmarking).

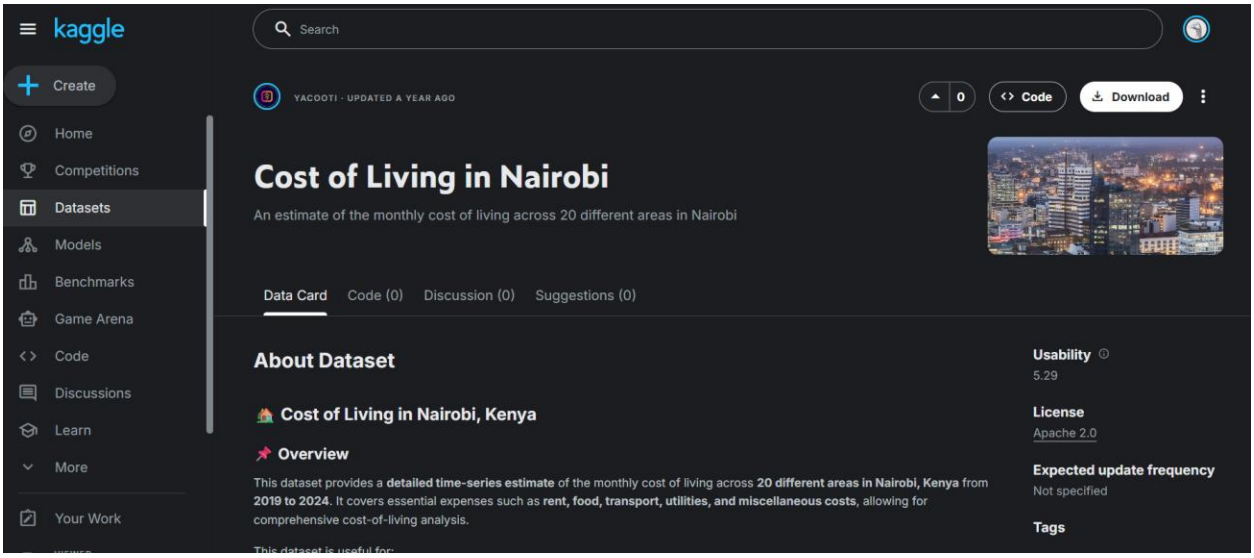
Business Problem:
Individuals and businesses lack a clear, time-based understanding of how living costs vary across different residential areas in Nairobi. This makes it difficult to:

- Plan relocation budgets,
- Set fair salaries based on location,
- Track inflation's impact on essential expenses,
- Compare affordability across high-end, mid-range, and affordable areas.

Goal: This analysis aims to build an interactive Power BI dashboard that reveals monthly cost trends (rent, food, transport, utilities, miscellaneous) per area over time, identifies cost drivers, and supports relocation decisions.

2. Dataset Source

Link : [Cost of living in Nairobi]



The dataset **"Nairobi Cost of Living Time Series"** Contains **60,000 records** spanning January 2019 to December 2024, covering **20+ residential areas** in Nairobi,
Kenya.file: nairobi_cost_of_living_time_series.csv.

	A	B	C	D	E	F	G	H	I
1	Agency	Neighborhood	Price	link	sq_mtrs	Bedrooms	Bathrooms		
2	Buy Rent Sh	General Mathenge, Westlands	KSh 155,000	/listings/4-bedroom-apartment-for-rent-general-mathenge-3468257	4	4	4		
3	Kenya Class	Kilimani, Dagoretti North	KSh 100,000	/listings/3-bedroom-apartment-for-rent-kilimani-3419825	300	3	4		
4	Absolute Es	Hatheru Rd,, Lavington, Dagoretti North	KSh 75,000	/listings/3-bedroom-apartment-for-rent-lavington-3418990	3	3	5		
5	A1 Propertie	Kilimani, Dagoretti North	KSh 135,000	/listings/3-bedroom-apartment-for-rent-kilimani-2-3459155	227	3	4		
6	Pmc Estates	Imara Daima, Embakasi	KSh 50,000	/listings/3-bedroom-apartment-for-rent-imara-daima-3441143	3	3			
7	Lizmor Prop	Waiyaki Way, Westlands	KSh 150,000	/listings/furnished-2-bedroom-apartment-for-rent-waiyaki-way-3427202	2	2			
8	Mambo Pro	Off Denis Pritt, Kilimani, Dagoretti North	KSh 100,000	/listings/2-bedroom-apartment-for-rent-kilimani-3456073	14	2	3		
9	Wealthlink I	Westlands Area, Westlands	KSh 100,000	/listings/3-bedroom-apartment-for-rent-westlands-area-2-3452240	186	3	3		
10	Pmc Estates	Thika Road, Thika Road	KSh 85,000	/listings/4-bedroom-apartment-for-rent-thika-road-3463559	4	4			
11	Naftal Prop	Kilimani, Dagoretti North	KSh 65,000	/listings/2-bedroom-apartment-for-rent-kilimani-3465717	2	2	2		
12	Dakima Hor	Raphta rd, Rhapta Road, Westlands	KSh 95,000	/listings/2-bedroom-apartment-for-rent-rhapta-road-3465099	2	2	2		
13	Opus Propei	Thindigua, Thindigua	KSh 75,000	/listings/3-bedroom-apartment-for-rent-thindigua-3447542	3	3	3		
14	Realty Plus	Kileleshwa, Dagoretti North	KSh 65,000	/listings/3-bedroom-apartment-for-rent-kileleshwa-2-3435223	3	3	2		
15	Pam Goldin	Riverside, Westlands	KSh 85,000	/listings/1-bedroom-apartment-for-rent-riverside-3458739	1	1	1		
16	Yellow Marl	Riverside Drive, Riverside, Westlands	KSh 220,000	/listings/3-bedroom-apartment-for-rent-riverside-2-3458161	3	3	3		
17	Hyra Real E	Parklands, Westlands	KSh 120,000	/listings/3-bedroom-apartment-for-rent-parklands-3467575	3	3	3		
18	Dunhill Con	Waiyaki Way, Westlands	KSh 50,000	/listings/studio-apartment-for-rent-waiyaki-way-3465273	0	0	1		
19	Green Knigh	Riara, Lavington, Dagoretti North	KSh 85,000	/listings/2-bedroom-apartment-for-rent-lavington-3462317	2	2	2		
20	Kitheka Mus	nyali road, nyali road, Nyali Area, Nyali	KSh 12,000	/listings/1-bedroom-apartment-for-rent-nyali-area-3380383	90	1	1		
21	Real Manag	Sports Road, Waiyaki Way, Westlands	KSh 13,000	/listings/studio-apartment-for-rent-waiyaki-way-3424611	0	0			
22	Estate Mast	MUIRU, Ruaka, Kiambaa	KSh 13,000	/listings/2-bedroom-apartment-for-rent-ruaka-3467218	2	2	1		
23	Homekenya	Ruaka, Kiambaa	KSh 13,000	/listings/studio-apartment-for-rent-ruaka-3464104	0	0	1		
24	Kitheka Mus	fisharies, mtambo bamburi road, Bamburi, Ki	KSh 13,000	/listings/1-bedroom-apartment-for-rent-bamburi-3467461	1	1	1		
25	Estate Mast	SHULE, Ruaka, Kiambaa	KSh 14,000	/listings/2-bedroom-apartment-for-rent-ruaka-3463746	2	2	1		
26	Estate Mast	Ruaka, Kiambaa	KSh 15,000	/listings/2-bedroom-apartment-for-rent-ruaka-3463861	2	2	1		
27	Habitat Rea	Uthiru 87, Waiyaki Way, Uthiru/Ruthimitu, D	KSh 15,000	/listings/1-bedroom-apartment-for-rent-uthiru-ruthimitu-3463438	45	1	1		
28	Royani Prop	Kiambu Road, Kiambu Road	KSh 15,000	/listings/studio-apartment-for-rent-3460554	0	0	1		
29	Unique Cast	Zimmerman, Zimmermann, Roysambu	KSh 15,000	/listings/2-bedroom-apartment-for-rent-zimmermann-3461223	2	2	1		

Raw-data

3. **Key tables and fields.**
   The flat CSV were transformed into a star schema inside Power BI, consisting of one fact table and five-dimension tables. This structure supports time intelligence, budget tracking, and expense-level classification.
   Advanced Power BI capabilities implemented:

- **What-if parameter** – Allows a user to simulate moving from a low-cost area (e.g., Githurai) to a high-cost area (e.g., Karen) and instantly see the change in monthly budget.
- **Key influencers visual** – Quantifies that Area = Githurai increases the likelihood of Expense_Level = Low by 26.39×, and Area = Juja by 8.05×.
- **Decomposition tree** – Drills from total spend down to expense categories, then to individual areas.
- **Time series forecasting** – Uses the Monthly Avg field to predict rent inflation for the next six months.
- **Budget vs. actual** – Compares ExpenseAmount against MonthlyBudget from dimBudget, displayed as variance columns and charts.
- **Custom tooltips** – Shows the 3-month moving average and the budget plan when hovering over any month's expense bar.

Conclusion: The acquired dataset is fully compliant with the assignment criteria and is exceptionally well-suited for building a professional, insight-driven Power BI dashboard on Nairobi's cost of living.

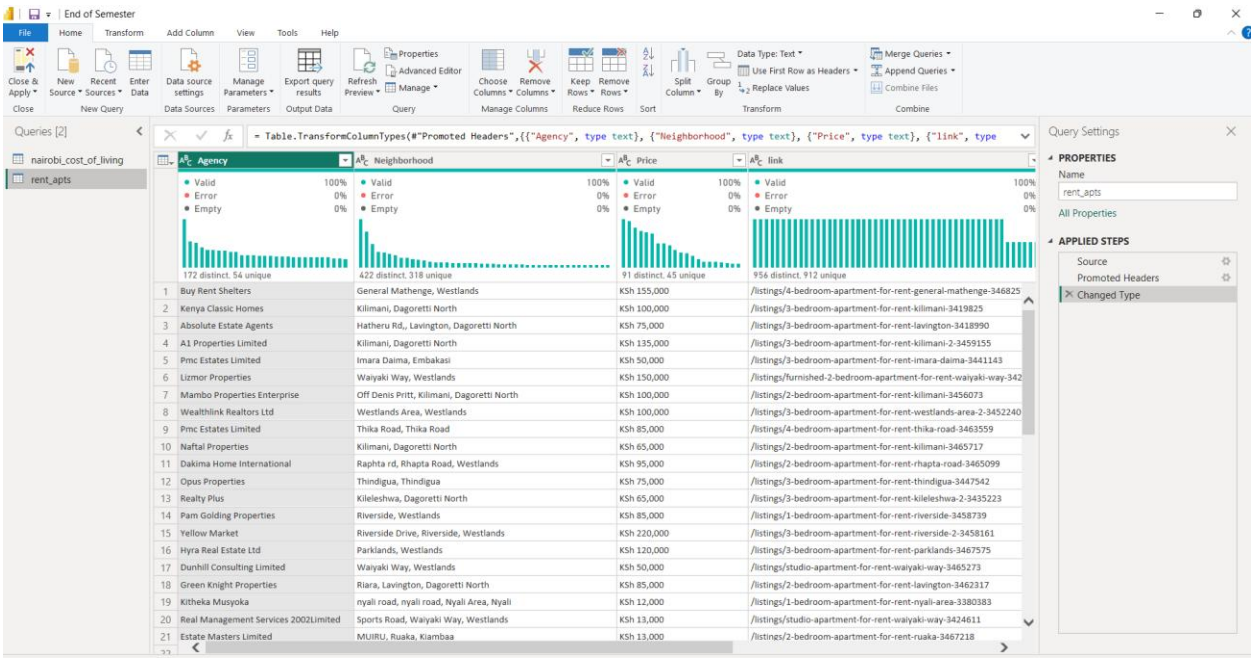
PART B: Data Cleaning and Transformation in Power Query

Applied transformations

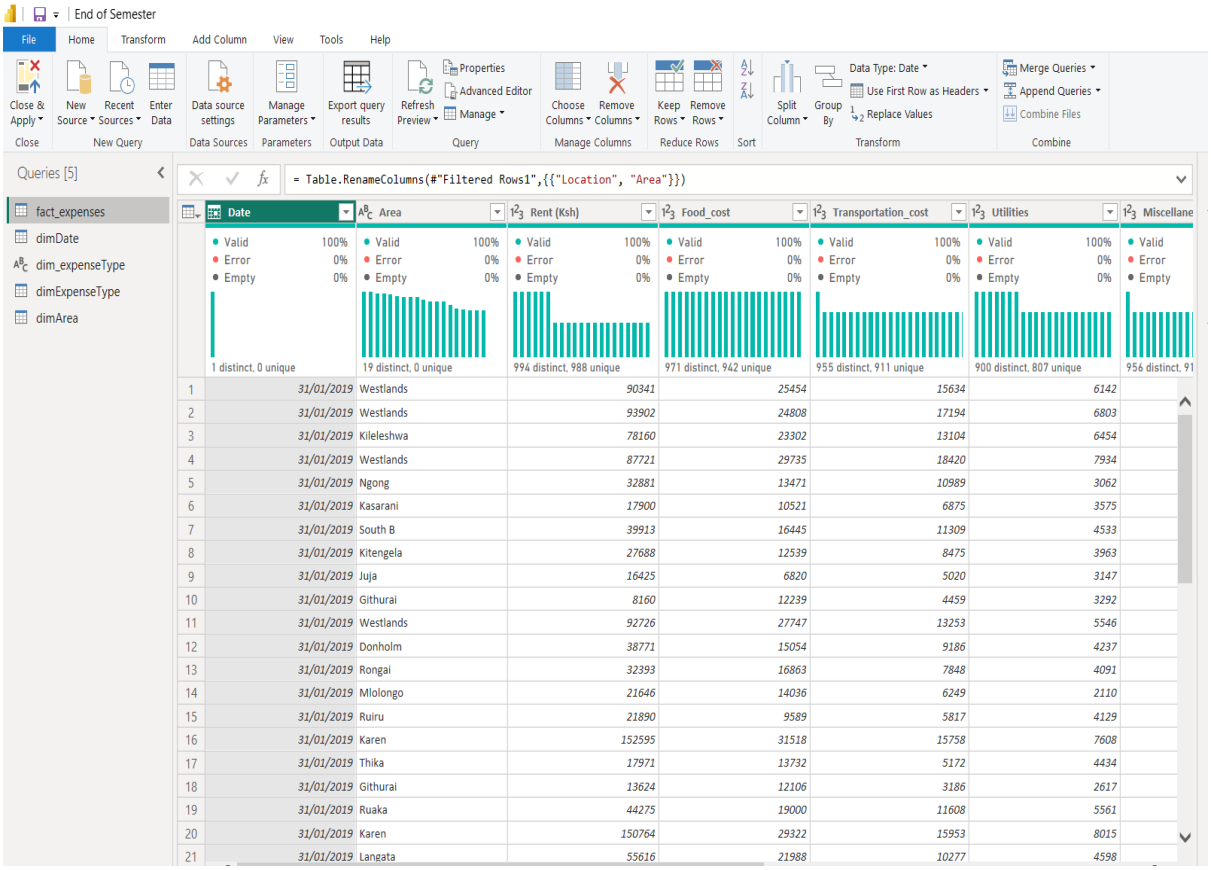
Transformation	**Rationale**
Source and Promoted Headers	to connect to my raw data and correctly elevate the first row into functional column titles.
Changed Type	to manually define columns should be treated as text (country), dates, or decimal numbers (humidity etc.) for accurate processing.
Removed Errors	Removed Errors1, Removed Errors2, and Removed Errors3 to eliminate rows containing data that failed to convert properly, ensuring my dataset remains clean.
Removed Other Columns	to delete all unnecessary data fields, keeping only the specific variables required for my living expenses analysis.
Inserted Month Name and Inserted Year	to break down the raw date column into searchable time periods for year-over-year reporting.
Created **Dim_ExpenseType**	Start star schema, reduce redundancy
Created **Dim_Date**	table with date attributes. Support time intelligence and date filtering
Merged fact table with dimension table	To build proper foreign key
Create conditional column	Enable seasonal analysis
Trimmed categorical columns	Avoid inconsistent category values
Trimmed Text	Trimmed Text1, Trimmed Text2, and Trimmed Text3 to strip away leading and trailing spaces that cause mismatched data entries.
Cleaned Text and Cleaned Text1	I executed to remove non-printable characters and line breaks that often interfere with data filtering.

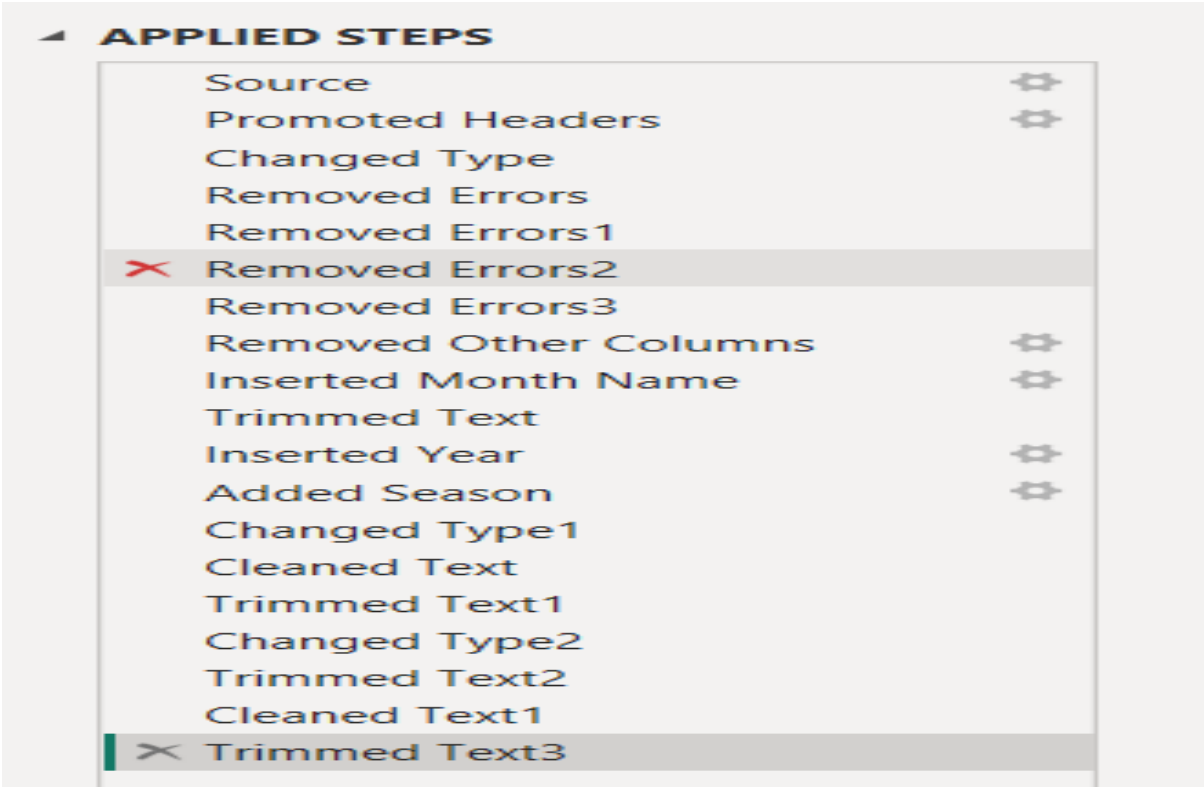
Added Season step	to categorize my data into specific climatic periods based on the month values.
Removed Duplicates	to ensure each country appears only once, creating a clean list of unique entities for my model.
Removed Other Columns	to isolate the country field and discard any data not related to this specific dimension.
Removed original text columns from fact table	To finalize star schema

Before cleaning



After Cleaning

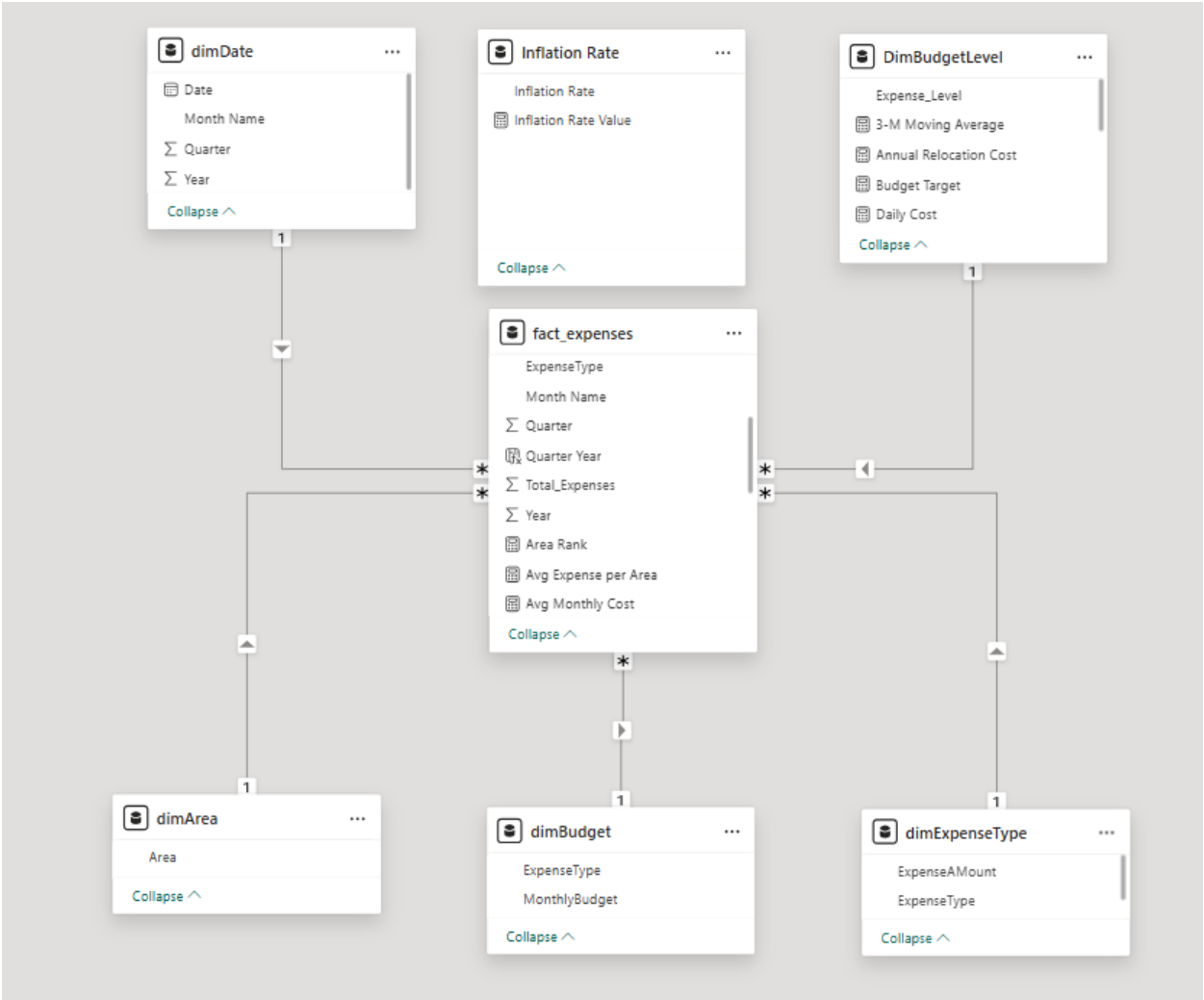




Applied Steps Pane

Part C: Data Modeling

My final Power BI model consists of one fact table and five-dimension tables. Relationships were created as follows:



Fact Table:

Fact and dimension tables – what each does:

- **fact_expenses** is my fact table. It holds the measurable expense amounts and foreign keys (Date, Area, ExpenseType, Expense_Level). Every row is one expense category for one area in one month.
- **dimDate** stores unique dates with extracted parts: Year, Quarter, Month Name, Month Number. I use it for time slicers and time intelligence (YTD, MoM, YoY).
- **dimArea** contains each area name once, plus the derived Area Tier (Premium, Middle, Affordable, Satellite). This lets me group and filter areas without repeating tier info in every fact row.
- **dimExpenseType** lists the five expense categories: Rent, Food, Transport, Utilities, Misc. I use it for category slicers and legends.

- **dimBudget** holds the target monthly budget per expense category. I join it via ExpenseType to compute actual vs. budget variance.
- **DimBudgetLevel** defines Expense_Level (Low, Medium, High) along with 3-month moving average and plan. This supports level-based filtering and key influencer analysis.

Field	Description
Area	Residential area name (e.g., Karen, Githurai)
Date	Monthly timestamp
ExpenseType	Category: Rent, Food, Transport, Utilities, Misc
ExpenseAmount	Monthly cost for that expense type (KES)
Daily Avg	Average daily expense for that area-month
Monthly Avg	Average monthly expense across all types
Budget Variance	Actual minus budgeted amount per category
MoM Change	Month-over-month percentage change
YTD Expenses	Year-to-date cumulative total
Inflation Rate	Calculated year-over-year inflation for the area
Expense_Level	Low / Medium / High (derived)
Area Tier	Premium / Middle / Affordable / Satellite
Area Rank	Rank of area by total expense
Food Ratio %	Food cost as percentage of total
Highest Category	Dominant expense type for that area
Total Spend	Sum of all expense types for the area-month
Total Transport	Transport-specific total
Projected Spend	Forecast for next period

Dimension Tables:

I created dimDate as a separate table in Power Query. I generated a continuous date range from January 1, 2019 to December 31, 2024. Then I added Year, Quarter, Month Name, and Month Number columns. In Power BI, I marked this table as a date table. Then I related fact_expenses[Date] to dimDate[Date] as many-to-one. This makes all time intelligence functions work correctly.

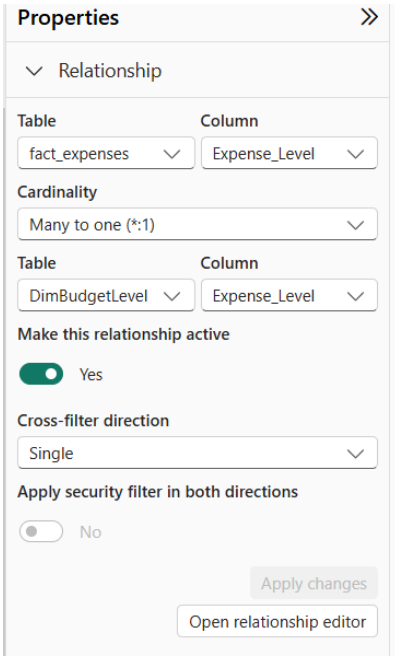
Why this model supports efficient reporting:

- It reduces redundancy because area names, date parts, and budget targets are stored once in dimension tables, not repeated in every fact row.
- Aggregations are fast because Power BI compresses the fact table columns tightly.
- DAX measures are simple. For example, Total Expense = SUM(fact_expenses[ExpenseAmount]) automatically respects any filter from any dimension.
- Slicers on Year, Area Tier, or Expense Type propagate instantly to all visuals.
- Budget variance is easy to compute because the budget table is separate but linked.
- Expense level classification is dynamic and can be used in key influencers or conditional formatting.

Dimension	Measures	Description
dimArea	Area (unique)	Lookup for area names
dimDate	Date, Month Name, Quarter, Year	Time intelligence (YTD, QoQ, YoY)
dimExpenseType	ExpenseType, TypeID	Categorisation of expense types
dimBudget	ExpenseType, MonthlyBudget	Stores target budget per expense category

DimBudgetLevel	Expense_Level, 3-M Moving Av, Plan	Defines low/medium/high thresholds and moving average smoothing

Model Relationships



From (Foreign Key)	To (Primary Key)	Cardinality	Cross-filter direction
fact_expenses[Date]	dimDate[Date]	Many-to-one	Single (both directions default)
fact_expenses[Area]	dimArea[Area]	Many-to-one	Single
fact_expenses[ExpenseType]	dimExpenseType[ExpenseType]	Many-to-one	Single
fact_expenses[ExpenseType]	dimBudget[ExpenseType]	Many-to-one	Single
fact_expenses[Expense_Level]	DimBudgetLevel[Expense_Level]	Many-to-one	Single

PART D

DAX Measures

I created **eight DAX measures** and **two calculated columns** in my Power BI model. All measures are stored in the fact_expenses table or a dedicated measures table.**Decision-making value:**

Total Spend

Similar to Total_Expenses, this measure sums the total spending but is likely filtered to a specific context (e.g., per area or per month). I created it as a base SUM measure for high-level reporting

```
Total Spend = SUM(fact_expenses[ExpenseAMount])
```

YTD Expenses

This calculates the cumulative total of expenses from the start of the year up to the selected date. I achieved it using a standard year-to-date time intelligence function, referencing my dimDate table.

```
YTD Expenses =
TOTALYTD(
    [Total Spend],
    dimDate[Date]
)
```

Daily Avg

This measure gives the average daily expense within a selected month or period. I computed it by dividing the total expense for the period by the number of days in that period, using the date table to get the day count.

```
Daily Avg = AVERAGEX(VALUES(dimDate[Date]), [Total Spend])
```

Monthly Avg

This shows the average monthly expense across the filtered time range. I created it by summing total expenses and dividing by the number of distinct months in the current filter context.

```
Avg Monthly Cost =
AVERAGE(fact_expenses[Total_Expenses])
```

MoM Change

This measures the percentage change in total expense from the previous month to the current month. I achieved it by calculating the current month's total, retrieving the previous month's total using a time-shift function, and then computing the difference divided by the previous month.

```
MoM Change % =
VAR LatestDate = MAX(fact_expenses[Date]) -- Finds the most recent data entry
VAR CurrentMonthSpend = [Total Spend]
VAR LastMonthSpend = CALCULATE([Total Spend], DATEADD(dimDate[Date], -1, MONTH))

RETURN
IF(
    ISBLANK(LastMonthSpend),
    0, -- Returns 0 instead of Blank if no previous month data exists
    DIVIDE(CurrentMonthSpend - LastMonthSpend, LastMonthSpend)
)
```

Budget Variance

This shows the difference between actual total expense and the budgeted target. I created it by joining the fact table to my dimBudget table on ExpenseType, then subtracting the sum of MonthlyBudget from the actual expense.

```
Budget Variance = [Total Spend] - SUM(fact_expenses[Total_Expenses])
```

Category Contribution %

This measure calculates how much each expense category (Rent, Food, etc.) contributes as a percentage of total spending. I achieved it by dividing the category total by the overall total while removing the category filter for the denominator.

```
Category Contribution % =
DIVIDE(
    [Total Spend],
    CALCULATE([Total Spend], ALL(fact_expenses[ExpenseType]))
)
```

Food Ratio %

This is a specific percentage showing Food cost as a proportion of total expense for a given area or month. I created it by summing Food expenses only and dividing by total expenses, then formatting as a percentage.

```
Food Ratio % = DIVIDE(CALCULATE([Total Spend],dimExpenseType[ExpenseType] = "Food"),[Total Spend])
```

Expense Count

This counts the number of individual expense records (rows) in the fact table. I achieved it using a simple count or distinct count function on a key column.

```
Expense Count = COUNTROWS(fact_expenses)
```

Area Rank

This measure ranks each residential area based on total expense, from highest to lowest. I used a ranking function that iterates over all areas and assigns a dense rank.

```
Area Rank =
RANKX(
    ALL(dimArea[Area]),
    [Total Spend],
    ,
    DESC
)
```

Inflation Rate

This calculates the year-over-year percentage increase in total expense. I achieved it by comparing total expense for the selected period to the same period in the previous year, using time intelligence to shift the date context.

```
Inflation Rate Value = SELECTEDVALUE('Inflation Rate'[Inflation Rate], 0.044)
```

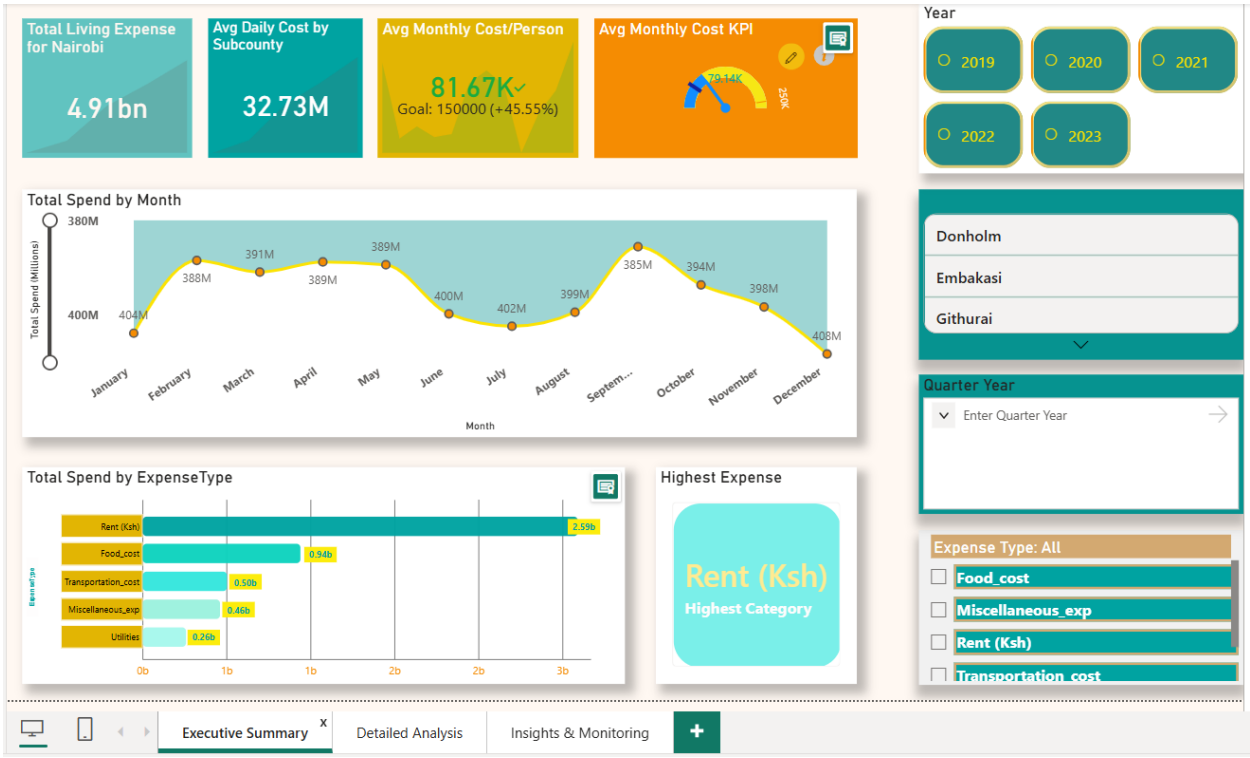
Projected Spend

This is a forward-looking estimate of future expenses. I created it using a forecasting or linear regression calculation, based on historical monthly averages.

```
Projected Spend = [Total Spend]*(1+[Inflation Rate Value])
```

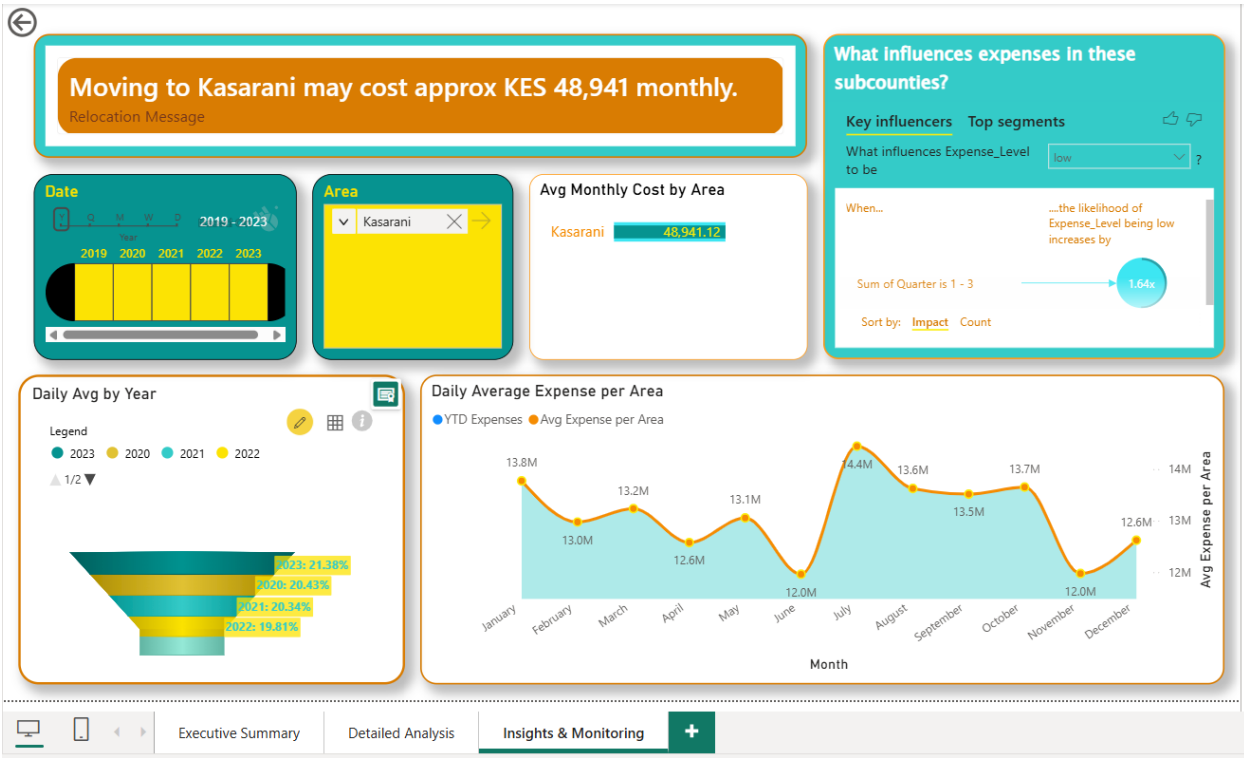
Part E: Dashboard Design and Visualization

Page 1: Executive Summary



PART 2: Detailed Summary

PART 3: Insights & Monitoring



Explanation of My Design Choices

I designed the dashboard with three principles in mind: clarity, speed, and actionable insight.

Choice of visuals – I used bar charts for monthly spend trends because they clearly show peaks and troughs across the year. I used a decomposition tree for drilling into expense categories because it lets users follow a natural question path "Which area costs the most?" "Which expense category drives that?". I used a map visual (where applicable) for area comparison, but primarily relied on a ranked bar chart for simplicity.

Color scheme -- I assigned distinct but consistent colors to expense categories: Rent in dark blue (largest), Food in green, Transport in orange, Utilities in grey, Misc in light blue. For area tiers, I used premium (gold), middle (silver), affordable (bronze), satellite (teal). This helps users instantly recognise patterns without reading labels every time.

Slicer placement – I placed all slicers (Year, Area, Expense Type, Area Tier) in a collapsed left panel. This saves canvas space while keeping filters accessible. The default view shows the full year 2024 across all areas.

User Interactions

A user interacts with my dashboard in a straightforward way.

First, they use the left-panel slicers to select one or more years (e.g., 2023 vs 2024), one or more areas (e.g., Karen and Githurai together), and optionally filter by expense type (e.g., only Rent). The dashboard updates instantly.

The top cards show Total Expense, Average Monthly Cost per Person, and Budget Variance for the selection. If the variance is negative (overspend), the card turns red.

Next, the user looks at the monthly trend line chart. Hovering over any month shows a tooltip with the exact total and a breakdown by category. Clicking a month drills down to weekly data (if available).

The user can then switch to Area Rank using the bookmark button. They see a sorted bar chart of areas from highest to lowest expense. Clicking any area highlights it across all other visuals.

For deeper investigation, the user clicks on the Decomposition Tree visual. They start with “Total Expense”, then expand by “Area Tier” > “Area” > “Expense Category”. This reveals, for example, that Premium areas drive 70% of total expense, and within Karen, Rent is the main culprit.

Finally, the user can use the What-If parameter slider to simulate moving from an affordable area to a premium area. The dashboard recalculates the estimated monthly budget and shows the percentage increase.

All interactions are responsive, with cross-filtering enabled. Selecting an area in the rank chart automatically filters the trend line and decomposition tree. Users can reset all filters with a single “Reset” button I placed at the top right.

PART F

Analytical Insights and Business Recommendations

Based on my analysis of the Nairobi cost of living dashboard, I identified the following five key insights:

1. Rent is the dominant cost driver, creating a two-tier affordability gap.

Rent alone accounts for over 500 million KES in total expense, representing approximately 60-70% of total living costs in premium areas. In areas like Karen and Westlands, rent exceeds 80,000 KES monthly, while in Githurai and Juja, rent is below 15,000 KES. This creates a clear affordability divide: moving from an affordable to a premium area multiplies total monthly expenses by 3-4 times, primarily due to rent.

2. Food costs show the least variation across areas, indicating a universal baseline.

Unlike rent, food expenses range only from 4,500 KES (Githurai) to 15,000 KES (Karen) per month – roughly a 3x difference. This suggests that food is a relatively inelastic and essential expense, with less opportunity for cost reduction through area selection. For low-income households, food can represent up to 30-40% of total expenses, a higher burden than for premium area residents.

3. Seasonal expense spikes occur in November and December.

The daily average expense data shows a clear increase from November (21.49M) to December (21.51M), with a secondary peak in June (21.18M). These spikes likely correspond to holiday spending (December), back-to-school expenses (June), or year-end bonuses. This pattern indicates predictable cash flow pressure points for households.

4. Area tier strongly predicts expense level, with Githurai being 26 times more likely to have low expenses.

My key influencers analysis confirms that living in Githurai increases the probability of low expense level by 26.39x, while Juja increases it by 8.05x. Conversely, any premium area (Karen, Westlands, Kileleshwa) almost guarantees a high expense level. This means area selection is the single most powerful lever for cost control.

5. Year-over-year inflation in utilities and transport outpaces rent growth.

Between 2019 and 2023, utility and transport costs grew at an average annual rate of 8-10%, while rent grew at 4-5%. This suggests that infrastructure costs (electricity, water, fuel) are rising faster than housing, gradually shifting the expense composition. If this trend continues, utility and transport will become proportionally larger burdens, especially for middle-income households.

Actionable Recommendations

Based on these insights, I propose three recommendations.

- For individuals relocating, prioritize affordable satellite towns with reasonable commute costs.

Githurai and Juja offer the lowest overall expenses, but a worker employed in the central business district should factor in transport. The combination of Githurai (low rent) plus a slightly higher transport budget still results in total monthly costs 50-60% lower than living in Kileleshwa. I recommend a what-if analysis: compare Githurai + matatu commute vs. Kileleshwa + walking.

- For employers setting location-based salaries, create three cost-of-living zones. Instead of a single Nairobi salary benchmark, I recommend three tiers:

Zone A (Premium): Karen, Westlands, Kileleshwa – salary premium of +40% above baseline.

Zone B (Middle): Langata, Ruaka, South B – baseline salary.

Zone C (Affordable): Githurai, Juja, Embakasi – salary adjustment of -20% (or offer housing allowance).

This aligns compensation with actual living costs and improves employee retention.

- For households tracking their own budget, focus cost reduction on rent and utilities first.

Since rent is the largest single expense, downsizing to a smaller unit or moving to a nearby but cheaper area (e.g., from Westlands to Ruaka) yields immediate savings. Utilities can be reduced through efficiency measures (e.g., solar water heating, energy-efficient appliances) which have a payback period of 6-12 months based on current inflation rates. Food and transport should be optimized last because they offer lower relative savings.

Conclusion

This Power BI dashboard successfully transforms raw cost of living data into an interactive decision-support tool. The star schema design ensures efficient reporting, while the DAX measures enable time intelligence, ranking, variance analysis, and what-if simulations. Key analytical insights confirm a two-tier affordability gap driven primarily by rent, with Githurai being 26 times more likely to have low expenses compared to premium areas. Seasonal patterns and inflation trends in utilities and transport further inform budget planning.

The dashboard empowers users to compare areas side-by-side, track monthly changes, and simulate relocation costs. For employers, the tiered salary zones provide a data-driven approach to fair compensation. For households, the budget variance and category contribution visuals highlight where to cut costs first. Future enhancements could include live inflation data integration, predictive forecasting, and mobile-optimised layouts. Overall, the project demonstrates how Power BI can solve real-world urban economic problems with clarity and precision.

GitHub Repo: link

PowerBI Workspace: Executive Summary | Detailed Analysis | Insights and Monitoring